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Research Article

From Exclusion to Participation: Digital Competencies and Participatory Finance for Rural Women in Tunisia

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Abstract

Financial inclusion of rural women entrepreneurs is conventionally related to gender issues; yet, it becomes more intriguing if we consider some explanatory factors, namely rurality, context and digital literacy. The present article aims at assessing the crucial drivers of women's adoption of crowdfunding in an emerging country. A quantitative study using structural equation modelling is adopted to test an extended UTAUT model of crowdfunding adoption. The sample comprises 250 female entrepreneurs active in Tunisian rural areas, a context marked by high female literacy yet persistent barriers to financial access. The results reveal that perceived usefulness, risk and trust influence rural women's crowdfunding adoption, with information and communication technology (ICT) knowledge acting as a relay for effective contextual influences. These findings imply that ICT knowledge contributes to financial inclusion mainly when it is tied to perceived usefulness, as a primary UTAUT factor, complemented by trust in platforms and risk management. Social influence becomes salient via ICT capacity building rather than through pressure. The study offers, besides the theoretical implications, insights for policymakers to unlock women's entrepreneurial potential and reinforce the principles of international mainstreaming related to human development, inclusiveness, gender equality and diversity.

Keywords

Crowdfunding, ICT knowledge, UTAUT, women rural entrepreneurship, gender, financial inclusion

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Introduction

The issue of women's entrance into the business landscape, both as employees and entrepreneurs, has been debated by academics as to how to apprehend such a new reality. Indeed, the proportion of female entrepreneurs continues to rise annually on a global scale. However, things have not been easy for women to succeed in ever since. They have been facing several socio-economic, institutional and financial barriers, depriving them of fully benefiting from their education, competencies, knowledge and personal motivation. The concepts of 'glass ceiling', 'discrimination' and 'stereotypes' are recurrent in studies on 'women at work' (Alhosseiny 2023; Xie et al. 2021; Zghal et al. 2006). Unfortunately, things have got 'harder' and 'complex' when it comes to considering the case of women in rural areas, as the latter are stigmatised by marginalisation, lack of access to education, resources and recognition as potential entrepreneurs (Ergün et al. 2025; Rezaei-Moghaddam 2025).

Literature review reveals that theories dealing with rurality and gender in entrepreneurship include a blend of contextual, social and institutional frameworks (de Guzman et al. 2020). These theories suggest capturing the complexity and interplay of local context, gender norms and systemic barriers that shape the trajectories of rural women entrepreneurs (Astner and Roos 2024; Gilani et al. 2024).

Certain disparities among regions of the world are noticed, such as the MENA region, where 'only 6 percent of total private equity and venture capital funding goes to female-led businesses' (Women Entrepreneurs Finance Initiative 2025). Some studies report the existence of structural and cultural barriers that deprive women of financing opportunities (Xie et al. 2021). In a study conducted in 57 countries, it was found that financial barriers and lack of experience constitute the primary challenges women entrepreneurs face in the 21st century, while lack of government support is cited in the last rank (Abdulla and Ahmad 2023).

However, we argue that even though the 'glass ceiling' persists, some factors have been enabling women to overcome these obstacles. One of the major factors is investing in education and particularly digital skills, allowing them to reach financial platforms such as crowdfunding. In fact, education provides women with essential capabilities, including digital literacy, persuasive communication for campaign design and a deeper understanding of financial technologies (Gilani et al. 2025a; Karami et al. 2025). In a study conducted across five developing countries (Bangladesh, Chile, Ghana, Brazil and the Philippines), the gender digital divide is found to vary in form and intensity, with Bangladesh being the most marginalised (Rashid 2016).

Even though information and communication technology (ICT) is considered a strategic resource that facilitates entrepreneurial innovation, some scholars underline the transformative potential of ICT in rural entrepreneurship (Gilani et al. 2025b). A recent study shows the persistence of inequalities in opportunities for digital education and connectivity for girls who may be deprived of devices in rural Tamil Nadu households (Balasundara et al. 2026). Besides, limited knowledge and experience are considered the foremost barriers to the adoption of digital technologies by farmers in Indonesia (Adana et al. 2026).

Based on this state of affairs, Chatterjee et al. (2020) note that the factors influencing ICT adoption by rural women are not comprehensively captured, and recurring obstacles to technology uptake may hinder their use of financing platforms. In particular, women's digital literacy is not sufficiently considered when examining their desire to use crowdfunding.

That is the case in most emerging countries witnessing high unemployment rates among women and, at the same time, low levels of financial support. Tunisian women continue to confront obstacles in obtaining financial and career prospects, even though their literacy rate hit 92.7% in 2020. This dichotomy triggers the question: To what extent does high ICT knowledge translate into greater financial inclusion when adopting crowdfunding by rural women entrepreneurs?

The study aims at investigating how and why rural women entrepreneurs in Tunisia tend to rely on collaborative platforms for financing, which will yield the detection of the key factors stimulating their adoption. It also suggests providing the actors of the ecosystem with a predictive model explaining women's intentions and real behaviours when dealing with the problem of financial access.

The article is structured into three main sections. The first presents a theoretical debate on the factors driving women's adoption of crowdfunding based on the UTAUT model. The second section is reserved for the presentation of the method used. The third discusses the results and concludes with the implications for theory and practice, the limitations and future research avenues.

Literature Review on Women's Behaviour Towards Crowdfunding Adoption

Our theoretical framework focuses on the examination of the UTAUT theory as a base to conceive our conceptual model explaining the key factors inciting rural women to adopt digital platforms for financing.

Crowdfunding as a Socially-constructed Financing Solution

Crowdfunding is a mechanism whereby individuals raise capital for various projects through online platforms, involving donations, loans or equity investments. The term 'crowdfunding' merges 'crowd' and 'funding', reflecting the mobilisation of a broad base of individuals through digital networks (Mollick 2014). It is characterised by the use of websites and social media to solicit contributions from the general public, transforming traditional fundraising into a decentralised and accessible process encompassing a broader spectrum of motivations and schemes (Ordanini et al. 2011).

Indeed, crowdfunding platforms make it possible for individuals to invest in projects, turning customers into stakeholders (Yáñez-Valdés and Guerrero 2024) in both civic and public initiatives (Thaker 2023). It offers an alternative funding mechanism for sustainable and appropriate technology projects, especially in contexts lacking institutional support (Wehnert and Beckmann 2021).

Globally, the crowdfunding ecosystem is rapidly expanding, with over 500 active platforms. Leaders like Kickstarter and Indiegogo dominate the market, particularly in countries such as the USA, France and China (International Finance Corporation 2022). Yet, some studies show that developing countries are still lagging behind due to the existence of certain sociocultural and economic specificities, such as those in Africa and South America (Abdulla and Ahmad 2023; Gilani et al. 2025a; Rashid 2016). As for the case of Tunisia, the International Finance Corporation (IFC)'s report indicates that crowdfunding is an emerging field with strong development potential, mainly with the milestones of the legal framework in 2020. Ten platforms are currently operational, including the national portal,[1] supporting a wide array of projects ranging from cultural to social impact entrepreneurial initiatives. Nevertheless, the sector in Tunisia still faces structural challenges, such as limited awareness, digital connectivity gaps and low levels of financial inclusion.

Theoretical Framework for Predicting the Adoption of Technological Innovations

Our theoretical framework intends to understand rural women's adoption of crowdfunding by evaluating factors that influence their behaviour based on the UTAUT model.

Unified Theory of Acceptance and Use of Technology: The Core Factors of Adoption

The focus is on the concept of adoption since it extends to the diffusion, intention and acceptance to emphasise the effective usage of technology. The mobilisation of the UTAUT goes to its generic nature, as it integrates key constructs from earlier technology adoption theories. According to Venkatesh et al. (2003), four factors are proven to determine behavioural intention and usage in technology adoption, as initially defined by Davis (1989): expected return, expected effort, social effects and favourable conditions. The author proves the moderating effects of age, gender, experience and the voluntary nature of usage. Since its introduction, researchers who have conducted empirical investigations on users' behaviours have frequently used UTAUT as a main framework in technology adoption.

Our research model incorporates perceived usefulness, social influence and facilitating conditions while extending the UTAUT framework by including perceived trust and risk, which are particularly relevant to the study of crowdfunding (Adana et al. 2026; Alalwan et al. 2017). Such reasoning relies on the following arguments.

First, perceived usefulness refers to the belief that using a particular system enhances job performance, increases efficiency and improves fundraising productivity (Adams et al. 1992; Davis 1989). Participatory financing allows entrepreneurs excluded from traditional credit channels to bypass bureaucratic processes and directly engage with potential funders.

Second, for some researchers, perceived usefulness significantly influences crowdfunding adoption (Kim and Jeon 2017; Li et al. 2018), whereas others report no significant effect (Moon and Hwang 2018).

Third, social influence implies that technology adoption is related to the degree to which individuals perceive that they should use a new system (Venkatesh et al. 2003). Findings reveal that it provides encouragement from reference groups (Rahi et al. 2018), contributes to project success (Ordanini et al. 2011) and shapes behavioural intentions (Yang et al. 2021). Therefore, trust and social influence become intertwined in the acceptance of digital financial technologies, substantiating their relevance in this study.

Finally, facilitating conditions concern 'the degree to which individuals believe that technical and organisational infrastructures are available to support system use' (Alalwan et al. 2017: 103). These conditions reflect the extent to which platforms are perceived as supportive environments, both technologically and institutionally. While platforms' feedback mechanisms and support systems may enhance user confidence, challenges such as poor data availability and insufficient reporting persist (Gerber et al. 2012). However, facilitating conditions do not always have a significant effect on users' intentions to adopt participatory finance (Moon and Hwang 2018), making them a crucial element in this study's conceptual model.

Extending the Technology Acceptance Model: New Perspective of Complementary Factors

Our reflection intends to deepen the analysis by integrating some variables that are susceptible to provide some insights into women's adoption of crowdfunding. Literature review depicts perceived trust, perceived risk and ICT knowledge as having great potential in apprehending women's behaviour regarding the adoption of funding platforms, thus increasing their chances of accessing financial support to realise their entrepreneurial projects (Geng and Xue 2023; Kalyani 2024).

Perceived trust relates to the characteristics of the platform, such as reliability, integrity and security (Adana et al. 2026; Zheng et al. 2016). It captures the confidence entrepreneurs and funders have in the platform's transparency, technical robustness and ethical standards. Li and Wang (2024) conceptualise trust as a dynamic process that develops over time and contributes to users' satisfaction beyond financial data.

Perceived risk is related to the potential for loss associated with using digital services (Featherman and Pavlou 2003). It covers several dimensions, such as financial, privacy, psychological and performance risks. However, research findings on the effect of perceived risk are inconclusive. While some studies provide evidence of the existence of its negative influence on behavioural intention (Lee et al. 2015; Moon and Hwang 2018), others argue that small contributions in crowdfunding mitigate perceived risks (Kim and Jeon 2017).

Therefore, it is relevant to assess the link between perceived trust and risk and digital knowledge in empowering rural women by enabling them to interact effectively with digital platforms despite the attitudinal and social challenges, as well as structural constraints, that may hinder effective inclusion (Adana et al. 2026; Geng and Xue 2023; Gilani et al. 2025b; Karami et al. 2025).

Hence, our study is grounded on the following main gaps in the literature. On the one hand, the focus seems to be mainly placed on urban areas, largely ignoring

rural specificities and gender issues in technology adoption (Eboagwu-ijah and Abwami 2024). On the other hand, even though crowdfunding platforms make financial inclusion possible, existing studies often neglect the socio-institutional factors in rural areas (Wu and Peng 2024; Pambudi et al. 2021). Crowdfunding's relevance for sustainable, locally tailored projects underscores the need to address these gaps (Adana et al. 2026; Awoke et al. 2025; Wehnert and Beckmann 2021).

Our study accounts for rurality's spatial constraints, gender dynamics and competency gaps, offering a nuanced lens for equitable digital inclusion. It is based on UTAUT theory and aims at examining the relevant intrinsic and extrinsic core motivating factors of behaviour in uncertain contexts related to the use of crowdfunding. It integrates some complementary factors inspired by the TAM model that explain the variation in crowdfunding adoption.

The analysis is based on the underlying hypothesis, declined into eight specific hypotheses related to the existence of a relationship between perceived usefulness, facilitating conditions, social impact, perceived risk and trust, and the adoption of crowdfunding by women in rural areas, with a mediating effect of ICT knowledge, as illustrated in Figure 1.

This conceptual rationale stems from the principle that posits that 'the behavioral intention to use a technology is the most proximal determinant of its actual acceptability' (Davis 1989: 985), whereas 'intention to use is the most direct measure of the acceptability of a technology by its potential users' (Venkatesh et al. 2003: 188). The constructs used expose the specific constraints, including technological limitations, gender-based stereotypes and geographic isolation. To evaluate the mediating role of ICT competencies in the adoption of crowdfunding, three items measuring digital knowledge levels were inspired by the scale proposed by Mat et al. (2021).

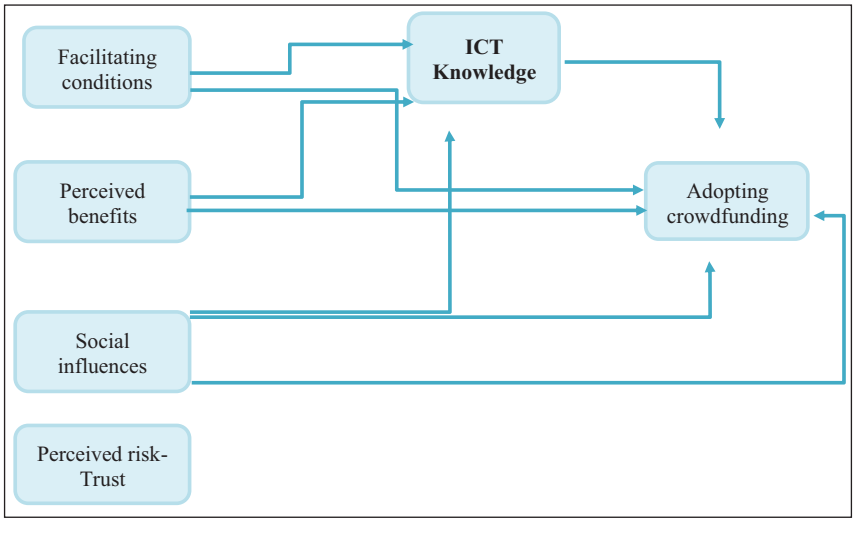


Figure 1. Conceptual Research Model.

Materials and Methodology

We conducted an empirical study aimed at providing evidence on the dynamics explaining how and why rural women behave in participatory funding opportunities. The present section describes the methodological choices engaged in the study. Then, we describe the methods of data collection and analysis, followed by a description of the sample.

Context of the Study

According to the latest available statistics, Tunisian rural women account for up to 1.4 million of the total population in rural areas, are dominant in agriculture (9% GDP, 16% jobs), yet are mostly unpaid and informal (43% workforce). The statistics of the World Bank indicate that 'Tunisia's ranking in terms of gender equality dropped from 90th to 124th out of a total of 153 countries between 2006 and 2020'. Although they constitute 67% of higher education graduates, 24.6% of them are active, whereas the unemployment rate reaches 35%. Even more, women's proportion of entrepreneurs is very low and barely reaches 4.07% (Verheijen 2020).

With regard to financial empowerment, the National Statistics Institute (NSI, 2024) reports that only 2.9% of Tunisian companies have a majority female capital. Women's share of microcredits fell despite sector growth, linked to informality and guarantees; only 2/7 of funding agencies (such as Enda and CFE) have women-specific strategies. Moreover, less than 20% of programme funds are accorded to women, whereas sectors financed are mostly in agriculture (29%).

It is worth noting that rural women in Tunisia have gained some attention both at the national and international levels through programmes aimed at their inclusion and empowerment. We may cite the project MOJA[2] for Adult Education Africa, aimed at providing agricultural skills, value chain development and climate-restricted tools and 'EmpowerHer' (Haddad 2018) initiative of the World Bank since 2016, with a focus on 'Technology for Women's Economic Empowerment'.

These facts underscore rural women's pivotal yet marginalised role, with ICT/crowdfunding potential for inclusion, knowing that the female unemployment rate has remained steady at 22% during the last decade (National Statistics Institute 2024). Based on this state of affairs, our model intends to provide actionable insights for policymakers to foster ICT training, platform localisation and gender-responsive regulations, ultimately bridging the rural gender gap in entrepreneurial finance (Women Entrepreneurs Finance Initiative 2025).

Data Collection and Analysis Techniques

Research Method and Sampling Strategy

We adopted a quantitative method since our research is both exploratory and explanatory. It was difficult to reach the respondents for interviews. Due to the lack of databases and statistics, we opted for the snowball sampling method. Snowball sampling is a non-probability, chain-referral sampling method in which existing participants recruit future participants from within their own networks. It

is especially used when the target population is hard to identify, access or list, a fact that is relevant for reaching dispersed rural entrepreneurs. Generally, in snowball sampling, the researcher starts with a small number of 'seed' participants who meet the inclusion criteria; after data collection with these seeds, each is asked to refer other eligible people, who in turn refer further participants, so the sample grows (Petersen and Valdez 2005). However, because it is non-probabilistic, it may lead to sampling bias and makes it inappropriate for statistical generalisation to the whole population. Results are therefore typically interpreted as context-specific and indicative.

The data collection process has yielded a sample of 300 rural women entrepreneurs, whose educational backgrounds ranged from high school to doctoral degrees, and who are active in projects requiring alternative financing such as crowdfunding. Email, Facebook and WhatsApp were among the social networks used for data collection. A total of 250 valid replies were used, with a response rate of 66.6%.

Data Analysis Methods

SmartPLS 3 was used to analyse the data collected and test the hypotheses. Partial least squares structural equation modelling (PLS-SEM) is especially well-suited for models with latent variables, small samples and non-normal data distribution (Hair et al. 2017). This approach enables both the exploration of relationships between the constructs and the evaluation of the predictive power of the model, making it well-suited to the objectives of our study.

Presentation of the Sample

The study sample consists of 250 women living in rural areas. They are predominantly young (84% aged between 18 and 40 years) and, interestingly, well-educated (80.8% have completed an undergraduate university degree). This population is marked by a high proportion of graduates (40% with 3 or 4 years, and 40.8% with 5 years) and a low representation of those with high school (13.2%) or doctoral (4%) levels of education. Even though these characteristics may hinder the generalisation of the results, they provide valuable insights into a young and educated demographic, potentially active in key sectors of the rural economy.

Results of Data Analysis

Using SmartPLS 3, the findings include the measurement model's validity and reliability and the structural model. The purpose is to test the assumptions and pinpoint the main variables affecting rural women's adoption of crowdfunding.

Assessment of the Measurement Model

The evaluation of the measurement model starts with convergent validity through average variance extracted (AVE). In accordance with the standard threshold

(AVE > 0.50), the multi-item constructs had AVEs ranging from 0.620 (social influence [SI]) to 0.790 (facilitating conditions [COND]), confirming satisfactory convergent validity, as shown in Table 1. Internal reliability is then assessed using composite reliability (CR) and Cronbach's alpha. The CR values are all above 0.70 and range from 0.830 (SI) to 0.919 (COND), while Cronbach's alphas range from 0.700 (SI) to 0.868 (COND), indicating good internal consistency of the constructs.

Besides, the outer loadings exceed 0.70, which supports item-level convergent validity (Hair et al. 2006). In fact, the contributions of the components to each element (load) for each structure vary from 0.714 to 0.903. As a result, our study model satisfies the constructs' convergent and internal validity criteria.

Discriminant validity is assessed using the Fornell–Larcker criterion (1981). Table 2 indicates that the square root of AVE exceeds the correlations below the diagonal for all constructs. This implies that the latent variables to which measurement indicators are allocated are the only ones that they can explain.

Table 1. Results of Internal Reliability Analysis.

Variables	Cronbach's Alpha	Rho-A	Reliability Composite	Average Variance Extracted (AVE)
Facilitating conditions (COND)	0.868	0.878	0.919	0.790
Perceived usefulness (PU)	0.757	0.761	0.860	0.673
Perceived risk (PR)	0.704	0.709	0.835	0.628
Social influences (SI)	0.700	0.704	0.830	0.620
Crowdfunding adoption (AC)	0.807	0.814	0.874	0.636
ICT knowledge	0.817	0.823	0.891	0.733

Source: Smart PLS output

Table 2. Results of Discriminant Validity Analysis.

	COND	SI	PR	PU	ICT Knowledge	AC
Facilitating conditions (COND)	0.889					
Social influences (SI)	0.699	0.787				
Perceived risk (PR)	0.759	0.753	0.826			
Perceived usefulness (PU)	0.726	0.758	0.772	0.820		
ICT knowledge	0.596	0.724	0.730	0.760	0.856	
Crowdfunding adoption (AC)	0.690	0.756	0.793	0.798	0.800	0.809

Source: Smart PLS output.

Table 3. Results of Hypothesis Testing.

Path	β Initial Sample (O)	Standard Deviation (STD)	*t*-Value	*p* Values	
COND → AC	0.026	0.071	0.362	.359	Rejected
SI → AC	0.092	0.066	1.392	.082	Rejected
PR/T → AC	0.350	0.079	4.419	.000***	Accepted
PU → AC	0.197	0.080	2.471	.007	Accepted
COND → ICT → AC	–0.037	0.026	1.427	.077	Rejected
SI → ICT → AC	0.086	0.035	2.472	.007	Accepted
PR → ICT → AC	0.096	0.033	2.918	.002	Accepted
PU → ICT → AC	0.132	0.035	3.762	.000***	Accepted

Notes: *** refer to the level of significance of p values.. AC: Crowdfunding adoption; COND: Facilitating conditions; ICT: ICT knowledge; PR: Perceived risk; PU: Perceived usefulness; SI: Social influences.

Assessment of the Structural Model

Structural equation modelling is used to test the significance of the direct relationships between the variables, based on path coefficients (β), *t*-statistics and *p* values obtained from the bootstrapping procedure. The results are compiled in Table 3.

Analysis of Direct Effects

The analysis reveals mixed results as to the direct effects of factors rural women entrepreneurs perceive as salient in their decision to adopt crowdfunding. More particularly, we may detect a certain scheme underlying such perceptions.

First, perceived risk/trust has a positive and significant effect on crowdfunding adoption ($\beta = 0.350$, $p = .000$). It is the most influential factor among all the variables considered in the analysis. The more trust women perceive, the more inclined they are to adopt crowdfunding. Second, perceived usefulness has a positive and significant influence on crowdfunding adoption ($\beta = 0.197$, $p = .007$). Women who see tangible benefits in crowdfunding are more likely to use it. Third, social influence in terms of peer pressure and cultural norms has no significant direct effect on women's decision to adopt participatory financing platforms ($\beta = 0.092$, $p = .082$). Finally, facilitating conditions do not have a direct effect on the adoption of crowdfunding ($\beta = 0.026$, $p = .359$). This means that rural women do not perceive these conditions related to infrastructure policies as sufficient or accessible enough to influence their decisions.

We may argue that, for our sample, the likelihood of adoption increases with the intrinsic factors related to the platforms in terms of trust and usefulness (H_3 and H_4 accepted). However, it seems that external factors do not have a great weight on how rural women deal with financing technologies (H_1 rejected, and H_2 is not significant).

Analysis of Indirect Effects

The analysis of indirect effects aims at assessing to what extent the initial scheme of direct effects is influenced by ICT knowledge developed by rural women entrepreneurs. The results, illustrated in Table 3, generate an amplified scheme of crowdfunding adoption shaped by a double-fold emergent path. On the one hand, the critical path (CR) of adoption aligns with the original one, with perceived usefulness being amplified indirectly via ICT knowledge in accordance with H_8 (β = 0.132, p = .000). Furthermore, perceived risk/trust moves to the second rank and indirectly reinforces crowdfunding adoption via ICT knowledge (H_7, β = 0.096, p = .002). This implies that women with a high level of ICT knowledge are inclined to view practical usefulness more clearly and manage risk through trust.

On the other hand, the initial external factors, namely SI (H_6) and COND (H_5), undergo a slight change in terms of their effect on crowdfunding adoption. More precisely, even though social influence does not have a direct effect, it becomes significant and acts indirectly via ICT knowledge to stimulate crowdfunding adoption (β = 0.086, p = .007). Interestingly, facilitating conditions remain insignificant both with and without mediation (β = –0.037, p = .077). This means that access to infrastructure alone is not enough to stimulate ICT knowledge creation and sharing.

Given these results, the overall explanatory power of the model was also assessed using R^2, which reached 0.794 for 'Adoption of Crowdfunding' and 0.655 for 'ICT Knowledge', with adjusted R^2 values of 0.786 and 0.644, respectively. This indicates that the model explains approximately 70% and 65% of the variance in the corresponding constructs.

In order to deepen the analysis for predictive and relevance adequacy, some relevant statistical indices are calculated, mainly f^2, Q^2 and goodness-of-fit (GOF). First, f^2, known as the effect size, measures each exogenous latent variable's contribution to the variance explained (R^2) in the endogenous constructs. Our results show that ICT knowledge has a moderate effect (f^2 = 0.174) on crowdfunding adoption, underscoring its crucial role in influencing participatory finance adoption. Similarly, perceived risk also has a moderate effect (f^2 = 0.161), highlighting its significance in the decision-making processes of rural women. In contrast, perceived usefulness and social influence unveil weak effects (f^2 = 0.051 and 0.013, respectively), indicating a less prominent but still relevant influence. As for the facilitating conditions variable, it shows a negligible effect (f^2 = 0.001), implying minimal influence.

Second, the Q^2 values for AC and ICT knowledge are 0.488 and 0.465, respectively, indicating moderate to strong predictive relevance.

Finally, the GOF index, calculated as the square root of the product of average AVE and average R^2, reaches 0.702. This exceeds the recommended threshold of 0.65, indicating excellent model fit and suggesting that the model achieves both strong measurement quality and explanatory power (Tenenhaus et al. 2005).

Discussion of the Results

This study examines the determinants of Tunisian rural women entrepreneurs' adoption of crowdfunding using a revisited UTAUT model. The results reveal

complex adoption mechanisms marked by interactions between psychosocial, technological and contextual factors, while highlighting paradoxes and dynamics specific to rural areas. They translate into higher perceived risks for both banks and women themselves when engaging with formal credit systems. As a result, many rural women remain trapped in low-productivity, informal activities despite growing levels of education and entrepreneurial intention.

Our results substantiate the assertion of Gancarczyk and Rodil-Marzábal (2022) that financial ecologies, as forms of territorial governance of financial services, have been profoundly transformed by financial technologies. Such dynamics underscore the importance of adapting financial services to local contexts, a key factor in promoting the inclusion of rural women through digital tools such as crowdfunding. In fact, the dynamics of participatory finance foster community engagement in project funding (Betsill et al. 2024; Cumming and Sewaid 2025). These mechanisms are highly linked to some Sustainable Development Goals (SDGs) related to gender equality (SDG 5), decent work (SDG 8), inclusive growth and reduced inequalities by targeting territorially marginalised populations in rural regions (SDG 10).

From a theoretical perspective, the study supports the UTAUT paradigm by showing that the adoption of crowdfunding is explained foremost by performance usefulness, unlike that of bakers in Korea, whose intention is not driven by usefulness (Moon and Hwang 2018). Our study shows that women perceive crowdfunding as a pragmatic, opportunity-enhancing tool that provides speed, flexibility and accessibility, especially where banking infrastructure is absent or limited, mainly in rural areas (Islam and Khan 2021; Kim and Jeon 2017). In other words, perceived usefulness reflects a rational calculus implying that when digital platforms offer real chances to secure capital without land titles or guarantors, the benefits clearly outweigh perceived costs.

Furthermore, the study shows a paradoxical result regarding the effect of perceived risk/trust, which serves as a catalyst and not as a barrier. Such a finding reinforces the primacy of perceived usefulness in explaining the intention and usage of crowdfunding platforms. Rural women tend to favour utility, leading to trust, which would compensate for risk. Recent studies in different contexts, such as Indonesia (Adana et al. 2026), Korea (Moon and Hwang 2018), and earlier in the USA (Featherman and Pavlou 2003; Gerber and Hui 2013), corroborate these trends by showing that trust in platforms is a major determinant of crowdfunding success, whereas perceived risks explain the extent to which digital solutions are adopted.

From a contextual perspective, the introduction of a crowdfunding law associated with the development of control mechanisms for platforms in Tunisia implies the existence of institutional support. The importance of such support lies not only in contributing to legitimising these tools in the eyes of women entrepreneurs but also in aligning with the building of effective and accountable institutions in reference to SDG 16.

Third, the mediating role of ICT knowledge is emerging as a key factor in assessing the relationship between perceived usefulness and trust/risk and crowdfunding adoption. This corroborates studies conducted in Indonesia (Adana et al. 2026) and Senegal (Adjanohoun and Agbanglanon 2022) , reporting that digital literacy increased the likelihood of fintech adoption by 53% and 40% of respondents, respectively. Perceived risk and trust are found to drive people's intention

to support crowdfunding platforms in Korea (Moon and Hwang 2018) and the USA (Gerber and Hui 2013).

However, inequalities in access to ICT, identified in the study on six developing African and South American countries (Rashid 2016), are reported as omnipresent in recent studies in Indonesia (Adana et al. 2026) and are alarming for young girls in rural Tamil Nadu (Balasundaram et al. 2026). These results underline the need for inclusive policies combining technical training and tackling structural barriers. It follows that, because of a social imbalance in the technological proliferation of ICTs, women's inclusion may vary with their position in some nations, resulting in a 'durable' and socially embedded gendered digital gap.

In addition, this study shows that social influence does not have a direct effect on crowdfunding adoption. Its effect through ICT knowledge reveals an essential community dynamic centred on learning experiences and gaining trust in digital tools. However, some studies report that social influence constitutes a main factor for crowdfunding adoption by American women (Geng et al. 2023) and also by Korean funders (Moon and Hwang 2018). We may deduce the emergence of a particular scheme of rural entrepreneurship that is based not only on geographic location but also on a strong connection to the local community. This observation aligns with some studies that show that women's groups use social networks to disseminate tutorials and create a collaborative learning ecosystem (Rumata and Sakinah 2020).

Finally, facilitating conditions in terms of technological infrastructure seem to lack a significant direct effect on women's adoption of crowdfunding contrary to the findings of Islam and Khan (2021). The results underline the importance of tailored strategies that go beyond infrastructure investment to include education, institutional reform and inclusive policy design. Some works advocate that digital transformation is accelerating worldwide and that the MENA economies must leverage these synergies to foster inclusive, resilient and innovation-led growth (Mohammed and Yacine 2025).

The mitigating results allow us to argue that social factors seem to lose their weight compared to more rational criteria such as perceived trust, risk and usefulness in crowdfunding adoption. Similarly, perceived benefits are necessary but insufficient to drive women to use ICT in their entrepreneurial activities. In essence, women may gain an advantage by investing in focused ICT knowledge in order to access financing, as the latter has been a major obstacle in promoting female entrepreneurship.

Rural entrepreneurship may, thus, be viewed as a learning process involving the gradual development of entrepreneurial skills, innovation and intercultural openness relevant to rural women in the adoption of crowdfunding. The idea is that crowdfunding becomes embedded within an ecosystem capable of empowering rural women and contributing to sustainable, inclusive development through the development of digital skills and familiarisation with new economic environments.

Conclusion

This study offers a framework for examining the relevant factors explaining women's usage of crowdfunding based on the UTAUT and knowledge-based theories. The findings reveal the prevalence of core factors of UTAUT, namely usefulness

and utility, and the emergence of some complementary factors such as perceived risk and trust. They provide evidence that, while ICT knowledge intervenes in linking these factors with crowdfunding adoption, social influence and facilitating conditions have a second-order effect.

The combined effect of gender, digital literacy and context, as exogenous factors, may be enhanced by attitudinal factors, namely perceived usefulness, risk and trust, which outweigh the effects of social influence and facilitating conditions. We are witnessing a multidimensional phenomenon that gains complexity, whereby tackling some contextual barriers generates new ones. In this regard, ICT and artificial intelligence (AI) knowledge evolves as a driving force to shatter the 'second glass ceiling' encountered by women entrepreneurs worldwide.

Theoretical Implications

This study enriches the UTAUT model by integrating complementary factors, mainly ICT knowledge and perceived trust and risk. It features the heterogeneity of adoption dynamics in rural areas, where technological and social factors interact in a non-linear fashion. This study significantly advances our knowledge of contextual factors affecting Tunisian women living in rural areas. The UTAUT model offers a pertinent framework for interventions to support participatory finance and rural women's economic empowerment, enhanced by the mediating role of ICT knowledge. From a contingent perspective, the issue of technology acceptance and diffusion may be tailored with gender and empowerment of specific categories in order to align with the diversity and inclusion principles of sustainable development.

Our results underline the emergence of a new paradigm in examining women's decisions on technology adoption that goes beyond the contextual view to embrace an attitudinal and cognitive view towards utility, risk and trust in fintech. In other words, the process of adoption may be further apprehended by integrating women's roles and personal strategies in overcoming barriers by investing in their knowledge favourably, making a difference in financial opportunities. From that point on, the issue of financial inclusion is to be inscribed within both the contextual view and the 'conventional' model of women entrepreneurs based on personal characteristics and strategies in a gendered, socially-embedded context.

From an epistemological viewpoint, our study suggests a paradigm surpassing the 'dichotomous and binary analysis' of gender issues embodied into themes such as 'access versus non access', 'opportunity versus barrier', 'rural versus urban', 'success versus failure' and 'context versus individual', while embracing a systemic and holistic contingent view.

Practical Implications for Stakeholders

The results call for several priorities to be taken by educational institutions, policymakers, NGOs and financial actors. In this regard, the government should prioritise raising educational standards to facilitate access to participatory financing

and optimise its advantages for rural women. Indeed, the focus on ICT education would enhance women's abilities and relevant knowledge in order to run campaigns efficiently and take full advantage of participatory funding opportunities, which is crucial for obtaining funds.

More specifically, women's digital education should be geared towards specific explicit and tacit knowledge with regard to useful applications such as crowdfunding. ICT knowledge programmes may include modules on digital security and the use of platforms, in partnership with local NGOs and institutions. Moreover, there is a need to enhance awareness-raising campaigns targeting the tangible benefits of crowdfunding, broadcast via community channels.

Furthermore, there is a need for tailored multisectoral public policies combining investment in infrastructure, subsidies for access to ICTs and quotas for the promotion of rural women in innovation programmes.

These measures should be inscribed in an effective ecosystem reinforced by the collaboration of several actors, namely entrepreneurs, governments, fintech and any interested parties, in order to transform crowdfunding into a lever for sustainable economic development for women's empowerment. Such an ecosystem calls for the institution of 'learning communities' that bring together all these initiatives in a structural manner.

The limitations of the study, notably its geographical focus and the absence of longitudinal data, call for extending the present research to unveil the evolution of adoption behaviour and the long-term impact of digital inclusion policies. Future research may be oriented towards a deeper understanding of the adoption of participatory finance in certain locations; other contextual and cultural factors might also be investigated. Comparative studies across nations and regions may also yield more information on the crucial factors affecting financial inclusion of rural women, including the impact of religion, educational attainment and the type of projects created by rural women.

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Notes

1. crowdfundingtunisia.tn
2. https://www.mojaafrica.net/

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